SO cross-cl pipeline: decision tree v3 — Bayesian search + profile scheme

left: search for moments with the evidence · right: profile scheme, every model at its own r

A · Bayesian search for moments (evidence, r free)

run BBPower without moments → $M_{0,r}$

is χ^2 good? (PTE > 0.01)
- No → Res_wr: residuals of $M_{0,r}$ all parameters fixed at their best fit (r too); test the d and s families of moments, each with its own β free
- Yes → is r compatible with zero at 1σ?
  - No → Res_wr: residuals of $M_{0,r}$ all parameters fixed at their best fit (r too); test the d and s families of moments, each with its own β free
  - Yes → Bayesian conclusion (posterior of $M_{0,r}$, r marginalised)

Res_wr: moment detected? (*) 0 outside the 68% HPD of an amplitude
- Yes → $M_{1,r}$: $M_{0,r}$ + a detected family, in order of residual evidence (**); full fit (all parameters free), pruned (***)
- No → Res_r0: residuals of $M_{0,0}$ (r = 0) all parameters fixed, own β free; test the families of moments d, s, dT, d2, s2, ds1, ds2

$M_{1,r}$: is $M_{1,r}$ preferred (****) over $M_{0,r}$? (ln B > 0)
- Yes → $M_{0,r}$ = $M_{1,r}$, start again
- No → Res_r0: moment detected? (*) 0 outside the 68% HPD of an amplitude
  - Yes → $M_{1,r}$: $M_{0,r}$ + the detected family with the highest residual evidence (**); full fit (all parameters free), pruned (***)
  - No → is r of $M_{0,r}$ compatible with zero at 3σ?
    - Yes → upper limit
    - No → r detection

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(*) all the amplitudes of the family are fitted (emcee); detected if 0 is outside the 68% HPD (highest posterior density interval) of at least one of them; only the amplitudes with 0 outside their HPD are kept

(**) residual evidence: ln Z of the residual fit of each detected family; it only sets the order in which the families are tried

(***) amplitudes with 0 in their 68% HPD (posterior of the full fit) removed and the model refitted, until all are detected; the pruned model is kept if ln B(pruned vs full) > 0 and is the one compared with $M_{0,r}$

(****) ln B within 2σ of the threshold (A) or between the two best models (B): PolyChord repeated with other seeds, mean of ln Z, up to 3 runs per fit

Bayesian conclusion (posterior of $M_{0,r}$, r marginalised)

then profile scheme

B · Profile scheme: every model at its own r

M* = final model of the tree ($M_{0,r}$)
candidates g: the families detected (among all tested) in its last Res_r0 (none if not reached or nothing detected)

profiles in r
M* → r* = its minimum
M* + g for each candidate g → r^*_g = its minimum

profile: r fixed on a grid, all the other parameters re-fitted (max likelihood)

compare all the models, each at its own r
M* at r*, M* + g at r^*_g for every candidate g:
local evidence (****) of each (its moment spectra free; β_d and β_s free in all, T_d only with dT)

final model = the one with the highest evidence
(M* itself if no candidate beats it)

profile of the final model in r
r_min = its minimum (r* or r^*_g);
posterior of r = exp(−Δχ²/2) × prior of the tree

other parameters
posterior conditional on r = r_min: emcee with r fixed, the rest free

0 within the central 3σ interval of the posterior of r?
- Yes → upper limit: r_95 of the posterior with r ≥ 0
- No → r detection: median and 68% of the posterior of r